

Software Test Plan (STP)

Team - 3 Hostel Management System

Project: Hostel Management System (HMS)

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1. Introduction

Purpose: This is our test plan for the Hostel Management System. It lays out what we're testing, how we plan to test it, and who's doing what, and when.

Scope: We're testing login and roles, student registration, block and room setup, room allocation, fees, complaints, the visitor log, the Admin dashboard, and search/filter. Reports and the CSV download on the Admin side are covered too. A few things are left out though: online payments, email/SMS, and support for more than one hostel. None of that is part of our SRS, so it's outside our scope.

References: Hostel Management System SRS (Team 3), Functional and Non-Functional Requirements (Team 3), Project Proposal (Team 3).

Definitions: HMS (Hostel Management System), SRN (Student Registration Number), Admin/Warden (same role, the one who manages the hostel), RTM (Requirements Traceability Matrix), TC (Test Case).

2. Test Items

- Login and Roles
- Student Registration
- Block and Room Setup
- Room Allocation
- Fee Management
- Complaint Management
- Visitor Log
- Admin Dashboard
- Search and Filter

- Reports and CSV download

3. Features to be Tested

Requirement IDs match the ones in our SRS.

- FR-1.1 to FR-1.5: Login, generic error for wrong credentials, correct dashboard for each role, Students blocked from Admin pages, logout
- FR-2.1 to FR-2.4: Registration, no duplicate SRN or email, student can edit own contact info but not SRN or role, Admin can edit any record
- FR-3.1 to FR-3.5: Adding blocks and rooms, unique room number within a block, a room or block with someone in it can't be deleted
- FR-4.1 to FR-4.8: Applying for a room, allotting and deallocating, no allotment to a full room, only one room per student, new room shows on the student's dashboard
- FR-5.1 to FR-5.6: Fee due and paid per term, balance calculated automatically, payment more than the balance is rejected, receipts, students see only their own fees
- FR-6.1 to FR-6.6: Raising a complaint, room and time added automatically, status goes Open, In Progress, Resolved, students can't change the status
- FR-7.1 to FR-7.5: Visitor entry and checkout, entry time taken from the server, list of visitors still inside, host student must have a room
- FR-8.1 to FR-8.5: Dashboard shows occupancy, dues, open complaints and pending applications, all taken live from the database
- FR-9.1 to FR-9.5: Searching students, filtering rooms and complaints, "no results found" message, students only search their own records
- NFR-1 Usability: applying for a room, raising a complaint and recording a payment each take at most two clicks
- NFR-2 Performance: room list, dashboard and search pages load within 2 seconds
- NFR-3 Security: hashed passwords, role checked on the server, students can't open other students' records by changing the URL
- NFR-4 Reliability: if two Admins allot the last bed at the same time, only one succeeds
- NFR-5 Scalability: still works with about 1000 students and 500 rooms
- NFR-6 Maintainability: every Django app has its own tests
- NFR-7 Portability: runs on Windows, Linux and macOS, works on Chrome and Firefox
- NFR-8 Availability: a crash shows a plain error page, not a stack trace

4. Features Not to be Tested

- Online payment gateway (fees are only recorded by Admin, not paid through the app)
- Email, SMS or any outside API (the SRS says there are none)
- Multiple hostels or colleges (the SRS assumes one hostel, one college)
- Special hardware (the SRS says only normal hardware is used)
- Database backups (SRS 6.2), since these are taken on the server and not inside the app

5. Test Approach / Strategy

Levels:

- Unit tests (each Django app: accounts, rooms, fees, complaints, visitors)
- Integration tests (views, models and database working together)
- System tests (full Student and Admin flows)
- Acceptance tests (UAT with the warden and hostel office staff)

Types:

- Functional testing (core features)
- Regression testing (re-running tests after every bug fix)
- Performance testing (2 second page load)
- Usability testing (two-click rule)
- Security testing
- Concurrency testing (two Admins allotting the last bed)

Entry Criteria: We need a stable build, test data ready, and the test environment set up before we start.

Exit Criteria: All planned test cases have run, no critical bugs are open, and everything in the RTM passes.

5.1 Security Validation

- Check that passwords are stored hashed (Django's PBKDF2) and never as plain text
- Log in as a Student and try to open Admin pages, including by typing the URL
- Change the ID in a URL to try to open another student's record
- Submit a form without a CSRF token and check that it fails
- Check that HTTPS is turned on in the deployment settings

6. Test Environment

Hardware: Just our own laptops and the college lab machines — nothing special needed.

Software: Python 3.10+, Django, SQLite while developing and PostgreSQL on the server, Chrome and Firefox.

Tools: Django test runner (unit and integration tests), Selenium (UI testing), Locust (load testing), GitHub Issues (bug tracking).

Test Data: Dummy student accounts, sample blocks and rooms, some complaints and visitor entries.

7. Test Schedule

Milestones:

- Test case design: 22-09-2026
- Environment setup: 24-09-2026
- Test execution start: 25-09-2026
- Test execution end: 05-10-2026
- UAT: 07-10-2026 to 10-10-2026

8. Test Deliverables

- Test Plan (this document)
- Test Cases (manual and automated)
- Test Scripts
- Test Data
- Test Execution Logs
- Bug Reports
- Test Summary Report

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	Shalik S	Prepare the plan, coordinate testing
Test Engineer	Shreyas Patil	Write and run test cases, log bugs
Developer	Shreyas Mohanty	Fix bugs and help find their cause

Role	Name	Responsibility
Product Owner	Shobhit Rachit Keshri	Check test results and give sign-off

10. Risks and Mitigation

Risk	Mitigation
Two Admins allotting the last bed at the same time could overfill a room	Write a separate test for this and run it before every release
SQLite and PostgreSQL may behave differently	Run the tests on PostgreSQL as well before release
We only have one server, so testing stops if it goes down	Keep a database backup and a local setup on our laptops
We are still new to writing Django tests	Test the first few modules together in pairs

11. Assumptions & Dependencies

- Only one hostel and one college use the system
- Admin sets up blocks and rooms before we test allocation
- Django's auth, ORM and admin site work as expected
- Test data is ready before testing starts

12. Suspension & Resumption Criteria

Suspend testing if:

- The test environment is down for more than 4 hours
- The build is too unstable (more than 30% of test cases are blocked)

Resume testing if:

- The blocking bugs are fixed
- The environment is working again

13. Test Case Management & Traceability

The RTM below maps each requirement from the SRS (Appendix C) to its test case.

Sl.	Req ID	Test Case	Expected Result
1	FR-1.2	TC-01	Wrong password shows a generic error with no hint about which field was wrong
2	FR-1.4	TC-02	Student opening an Admin URL directly is blocked
3	FR-2.2	TC-03	Registering with an SRN that is already used is rejected
4	FR-3.3	TC-04	Two rooms with the same number in the same block are rejected by the database
5	FR-4.4	TC-05	Allotting a full room is rejected
6	FR-4.5	TC-06	Allotting a second room to a student who already has one is rejected
7	FR-5.4	TC-07	A payment bigger than the balance owed is rejected
8	FR-6.3	TC-08	A new complaint has status Open
9	FR-6.5	TC-09	Student trying to resolve their own complaint is rejected
10	FR-7.5	TC-10	Visitor logged against a student with no room is rejected
11	FR-9.4	TC-11	Search with no matches shows an empty list with a message, not an error
12	NFR-3	TC-12	Student changing the ID in the URL to see another student's record is blocked
13	NFR-4	TC-13	Two allotments for the last bed at the same time: one succeeds, one is rejected

14. Test Metrics & Reporting

Metrics collected:

- % of test cases run
- % passed/failed
- Number of bugs per module
- How long bugs stay open
- Requirement coverage

Reports:

- Daily test status

- Final Test Summary Report

15. Approvals

Role	Name	Signature / Date
QA Lead	Shalik S	
Developer	Shreyas Mohanty	
Product Owner	Shobhit Rachit Keshri	